Discrete Choice Models

Cheng Chou

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1 Introduction

- It is common that we want to explain a binary choice in terms of some explanatory variables.
- Examples include labor force participation, education, insurance, migration, retirement et al.
- In this cases y can take two values 0 and 1. Given a vector of x, we are interested in

$$p(x) = \Pr(y = 1 \mid x) = \mathrm{E}(y \mid x).$$

  $p(x)$ is usually called CCP (conditional choice probability). Also, note $\mathrm{Var}(y|x) = p(x)(1 - p(x))$ (heteroskedasticity).

- Similar to linear regression, we are interested in the partial effect,

$$\frac{\partial \mathrm{E}(y \mid x)}{\partial x_j} = \frac{\partial p(x)}{\partial x_j}$$

- For discrete x_K such as treatment dummy variable (received job training), the "partial effect" becomes

$$p(x_1, \ldots, x_{K-1}, x_K = 1) - p(x_1, \ldots, x_{K-1}, x_K = 0)$$

- Unlike linear regression model, the partial effect **depends** on the value of x. This is an important feature in nearly all nonlinear models.

- The fact that the partial effect varies across of x makes nonlinear model more realistic.

- We usually use the following summary statistics

$$\begin{aligned} \text{Average partial effect (APE):} \quad & \mathrm{E}\left(\frac{\partial p(x)}{\partial x_j}\right) \\ \text{Partial effect at the average (PEA):} \quad & \frac{\partial p(\mathrm{E}(x))}{\partial x_j} \end{aligned}$$

- For binary x_K, the APE is

$$\mathrm{E}_{x_{(K)}}(p(x_{(K)}, 1) - p(x_{(K)}, 0)),$$

  where $x_{(K)} = (x_1, \ldots, x_{K-1})'$.

- In nonlinear models, the APE and PEA can be very different. APE is preferred, because $\mathrm{E}(x)$ might not even correspond to an individual in population.

2 Linear probability model

Linear probability model

- If we ignore the binary choice feature of y and view it as a dependent variable, we specify $\mathrm{E}(y \mid x)$ as a linear model,

$$\Pr(y = 1 \mid x) = \mathrm{E}(y \mid x) = \beta_1 + \beta_2 x_2 + \cdots + \beta_K x_K = x'\beta.$$

- Obviously, the OLS consistently estimate β. Indeed, it is also unbiased.

- It is inefficient though by its nature $\mathrm{Var}(y|x) = x'\beta(1 - x'\beta)$. This means we should use the heteroskedasticity robust variance estimator for inference.

- Linear probability model (LPM) is a good starting point for the ease of interpreting the coefficients.

Drawbacks of LPM

- The fitted values, $\hat{y}_i = x_i'\hat{\beta}$, might fall outside the unit interval. This is odd because $x_i'\hat{\beta}$ is the estimate of the probability $p(x_i)$.

- It is difficult to impose the diminishing effects of the x_j on the $p(x)$. For example, $p(x) = x'\beta$ means that $\partial p(x)/\partial x_j = \beta_j$. The partial effect does not diminish in magnitude.

- The estimation is inefficient due to the nature heteroskedasticity. Though we know $\mathrm{Var}(y|x) = x'\beta(1-x'\beta)$, and weighted least square (WLS) with the optimal weighting is
$$\min_b \sum_{i=1}^n (y_i - x'b)^2 / \mathrm{Var}(y_i|x_i).$$
In practice, since we cannot guarantee that $\hat{p}_i = x_i'\hat{\beta} \in (0,1)$, $\hat{p}_i(1-\hat{p}_i)$ might be negative.

- WLS is hardly attractive since we can always use the heteroskedasticity robust variance.

3 Logit and probit models

Index model

- Logit and probit belong to the general index model:
$$\Pr(y = 1 \mid x) = G(x'\beta),$$
for some $G : \mathbb{R} \mapsto (0,1)$. In statistics literature, generalized linear model in particular, G is called "link function".

- In most cases, including logit and probit, G is a cumulative distribution function for a continuous random variable with density g.

- Logit: $G(t) = \exp(t)/(1+\exp(t))$ and Probit: $G(t) = \Phi(t)$, where Φ is the CDF of the standard normal distribution.

Estimation with the maximum likelihood estimation

- Maximum likelihood estimation (MLE) is used to estimate many models, including logit and probit. It can be understood from the M estimation framework.
- Let $z_i = (y_i, x_i)$ denote a random draw. Suppose we specify the conditional density $f(y_i|x_i;\theta)$ up to a finitely dimensional vector of parameters $\theta \in \Theta$. The true value of θ is denoted by θ_0.
- Note that density is well defined for both continuous and discrete y.
- Probit example: $f(y = 1 \mid x;\theta) = \Phi(x'\theta)$ and $f(y = 0 \mid x;\theta) = 1 - \Phi(x'\theta)$.
- Poisson regression: $y \in \{0, 1, 2, \ldots\}$ and $\mathrm{E}(y \mid x) = m(x, \theta)$. Then

$$f(y \mid x;\theta) = \exp(-m(x,\theta))\frac{m(x,\theta)^y}{y!}.$$

Consistency of MLE

- Given $f(y_i|x_i;\theta)$ and *random* samples (y_i, x_i) for $i = 1, \ldots, n$, the probability of observing $(y_1, \ldots, y_n)$ given $(x_1, \ldots, x_n)$ given θ is

$$f(y_1, \ldots, y_n|x_1, \ldots, x_n;\theta) = \prod_{i=1}^{n} f(y_i|x_i;\theta)$$

- A natural estimator $\hat{\theta}_n$ of θ_0 is the θ that maximizes $f(y_1, \ldots, y_n|x_1, \ldots, x_n;\theta)$. Equivalently, $\hat{\theta}_n$ minimizes

$$-\log f(y_1, \ldots, y_n|x_1, \ldots, x_n;\theta) = \sum_{i=1}^{n} -\log f(y_i|x_i;\theta).$$

- This set up fits the M estimation framework by letting $q(z_i, \theta) = -\log f(y_i|x_i;\theta)$.
- To show the consistency, we need to show that θ_0 is the unique minimizer of

$$Q(\theta) = -\mathrm{E}(q(z_i,\theta)) = -\mathrm{E}(\log f(y_i|x_i;\theta)).$$

  This follows from the well known Kullback-Leibler information inequality:

$$\mathrm{E}(\log f(y_i|x_i;\theta_0) \mid x_i) \geq \mathrm{E}(\log f(y_i|x_i;\theta) \mid x_i),$$

  for all $\theta \in \Theta$. The equality holds if and only if $f(y|x;\theta) = f(y|x;\theta_0)$. So that θ_0 minimizes $Q(\theta)$.

- For identification, we assume that it is the unique minimizer.
- The Kullback-Leibler information inequality can be proved by Jensen's inequality. Because log is concave, Jensen's inequality implies that

$$\mathrm{E}(\log(f(z_i;\theta)/f(z_i;\theta_0))) \geq \log \mathrm{E}(f(z_i;\theta)/f(z_i;\theta_0)) = \log(1) = 0.$$

  In the above display, view $f(z_i;\theta)/f(z_i;\theta_0)$ as a single argument of log function.

Asymptotic normality of MLE

- We simply apply the M estimation results to derive the asymptotic distribution of the MLE.
- We have the score function,

$$s_i(\theta) = s(z_i,\theta) = -\nabla_\theta \log f(y_i|x_i;\theta),$$

  and the Hessian

$$H_i(\theta) = H(z_i,\theta) = \nabla_\theta s_i(\theta).$$

- We know $\mathrm{E}(s_i(\theta_0)) = 0$ by the FOC of $\min \mathrm{E}(\log f(y_i|x_i;\theta))$.
- Then the A and B matrices are

$$A = \mathrm{E}(H_i(\theta_0)), \quad B = \mathrm{E}(s_i(\theta_0)s_i(\theta_0)').$$

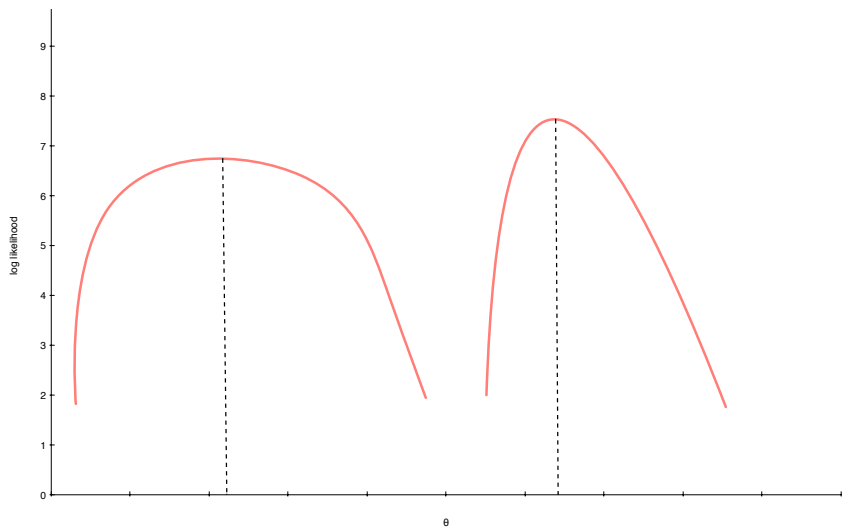
  Matrix B is called *Fisher information* matrix.

"Fisher information"

- Why is $B = \mathrm{E}(s_i(\theta_0)s_i(\theta_0)')$ called "information"?
- For scalar θ, Fisher information $\mathrm{E}(s_i(\theta_0)s_i(\theta_0)')$ reads

$$\mathrm{E}\left(\left(\frac{\partial \log f(z_i;\theta_0)}{\partial\theta}\right)^2\right).$$

- If this "information" is larger, it would be "easier" to estimate θ_0. Why? See the graph. In this sense, we have more information about θ_0.



- Before jumping to use $n^{-1}A^{-1}BA^{-1}$ as the variance formula, we have the following *information matrix equality*:

$$\mathrm{E}(H_i(\theta_0) \mid x_i) = \mathrm{E}(s_i(\theta_0)s_i(\theta_0)' \mid x_i).$$

$$\mathrm{E}(H_i) = \mathrm{E}(s_i(\theta_0)s_i(\theta_0)') \Rightarrow A = B.$$

- To see this, consider scalar θ. We have

$$s_i(\theta_0) = \frac{f'_\theta(z_i, \theta_0)}{f(z_i, \theta_0)}.$$

Hence

$$H_i = \frac{f''_\theta(z_i, \theta_0)}{f(z_i, \theta_0)} - \left(\frac{f'_\theta(z_i, \theta_0)}{f(z_i, \theta_0)}\right)^2.$$

The equality can be proven by noting that

$$\mathrm{E}\left(\frac{f''_\theta(z_i, \theta_0)}{f(z_i, \theta_0)}\right) = \frac{\partial^2}{\partial\theta^2}\,\mathrm{E}\left(\frac{f(z_i;\theta)}{f(z_i;\theta_0)}\right)\bigg|_{\theta=\theta_0} = \frac{\partial^2}{\partial\theta^2}\int f(z,\theta)\,\mathrm{d}\,z\bigg|_{\theta=\theta_0} = 0.$$

- Thus, the asymptotic variance of MLE is $n^{-1}A^{-1} = n^{-1}B^{-1}$.

- So we have two formulas for the asymptotic variance of MLE,

$$\left(\sum_{i=1}^n H_i(\hat\theta_n)\right)^{-1}, \quad \left(\sum_{i=1}^n s_i(\hat\theta_n)s_i(\hat\theta_n)'\right)^{-1}.$$

- In practice, the Hessian formula is preferred for its smaller finite sample bias.

- In the above discussion, we assume the model $f(y|x,\theta)$ is correctly specified. If the model was misspecified, the MLE is called quasi-MLE (QMLE). White (1982, Maximum Likelihood Estimation of Misspecified Models, Econometrica) shows that even when the model is misspecified, MLE still makes sense, because the estimated distribution is "closest" to the true distribution among the all possible distributions satisfying the wrong specification.

Likelihood ratio (LR) test

- Likelihood ratio test is simply

$$LR = 2(\mathcal{L}_{ur} - \mathcal{L}_r) = 2\left(\sum_{i=1}^{n} \ell_i(\hat{\theta}_n) - \sum_{i=1}^{n} \ell_i(\tilde{\theta}_n)\right) \to_d \chi^2_q.$$

  Here q is the number of constraints.

- LR test makes more sense in the MLE due to the information matrix equality ($A = B$).

- Wald and LM test can also be applied. The Wald statistic is not invariant to how the model is parameterized because it was based on the delta method.

MLE for index model

- Given random samples $\{y_i, x_i : i = 1, \ldots, n\}$, the log likelihood is

$$\ell_i(\beta) = (1 - y_i)\log(1 - G(x_i'\beta)) + y_i \log G(x_i'\beta)$$

- The asymptotic variance is

$$\left(\sum_{i=1}^{n} \frac{g(x_i'\hat{\beta})^2}{G(x_i'\hat{\beta})(1 - G(x_i'\hat{\beta}))} x_i x_i'\right)^{-1},$$

  where

$$g(z) = \phi(z) \quad \text{for probit}$$
$$g(z) = \exp(z)/(1 + \exp(z))^2 \quad \text{for logit}$$

- Testing multiple, nonlinear hypotheses about β is straightforward with the Wald or LR statistics.

- In Stata, Wald test can be done by `test` command. Stata reports the log likelihood value. So LR test can be easily done by hands.

Estimating partial effects: what does β tell us?

- The partial effects are
$$\frac{\partial p(x)}{\partial x_j} = \beta_j g(x'\beta).$$
Because $g(\cdot) > 0$, the sign of β_j gives the direction of the partial effect. But the magnitude depends on $g(x'\beta)$.

- Because the scale factor $g(x'\beta)$ is the same for each $\beta_1, \ldots, \beta_K$, β_j/β_h is useful:
$$\frac{\partial p(x)/\partial x_j}{\partial p(x)/\partial x_h} = \frac{\beta_j g(x'\beta)}{\beta_h g(x'\beta)} = \frac{\beta_j}{\beta_h}.$$

- No simple relationship exists for discrete variables or changes, because we cannot take β_K out of $G(x'\beta)$:
$$G\left(x'_{(K)}\beta_{(K)} + \beta_K \cdot 1\right) - G\left(x'_{(K)}\beta_{(K)} + \beta_K \cdot 0\right).$$

PEA and APE

- PEA estimate:
$$\widehat{PEA}_j = \hat{\beta}_j g(\bar{x}'\hat{\beta}).$$

- $\bar{x}$ does not make sense sometimes, e.g. if x includes both age and age^2, then it is unlikely to find an individual corresponding to $(\overline{age}, \overline{age^2})$, since $\mathrm{E}(age)^2 \geq \mathrm{E}(age^2)$.

- APE estimate:
$$\widehat{APE}_j = \hat{\beta}_j \left(n^{-1} \sum_{i=1}^{n} g(x_i'\hat{\beta})\right) \to_p \mathrm{E}(\beta_j g(x'\beta)).$$

- There is no certain relationship between PEA ($\beta_j g(\mathrm{E}(x'\beta))$) and APE ($\beta_j\, \mathrm{E}(g(x'\beta))$). For probit, $g = \phi$, which the pdf of standard normal distribution. We know that $\phi''(x) = (x^2 - 1)\phi(x)$. That is if $x \in [-1, 1]$, ϕ is concave. Otherwise, ϕ is convex.

- It is easy to derive the variance of $\widehat{PEA}_j$ by the delta method.

- It makes no sense to compare the magnitudes of coefficients across probit, logit, LPM. Comparing APE makes more sense.

Stata practice

- In Stata, `probit` and `logit` runs probit and logit estimation.

- `margeff` reports the APE, `mfx` shows the PEA. See slides week8a of EC2011.

Stata sometimes does not do the job

- If the index form $x'\beta = \beta_1 + \beta_2 x_2 + \cdots + \beta_K x_K$, where $x_2, \ldots, x_K$ are distinct variables, the APE reported by Stata makes sense.
- Consider the following specification,

$$p(x) = G(\beta_0 + \beta_1 x_1 + \beta_2 x_1^2 + \beta_3 \log(x_2)) = G(z'\beta),$$

  where $z = (1, x_1, x_1^2, \log(x_2))'$. You cannot use `margeff` to obtain APE, because Stata does not know functional relationships among the components of z.
- We have

$$\begin{aligned} \frac{\partial p(z)}{\partial x_1} &= (\beta_1 + 2\beta_2 x_1) g(x'\beta) \\ \frac{\partial p(z)}{\partial x_2} &= \beta_3 x_2^{-1} g(x'\beta) \\ \frac{\partial \log p(z)}{\partial \log x_2} &= \frac{\beta_3 g(z'\beta)}{G(z'\beta)}. \end{aligned}$$

- It would be easier to compute these numbers with Matlab or R.

4 Specification issues

Latent variable formulation

- The probit model can also be understood from the following form,

$$y_i^* = x_i'\beta + u_i, \quad u_i \sim \mathcal{N}(0,1), \quad y_i = 1(y_i^* \geq 0)$$

  Here y_i^* is called latent variable.
- You may wonder why do we need to let $\mathrm{Var}(u_1) = 1$.
- We need to normalize $\mathrm{Var}(u_1) = 1$ or some constant. To see this,

$$\text{Model I: } y_i^* = x_i'\beta + u_i, \quad u_i \sim \mathcal{N}(0,1), \quad y_i = 1(y_i^* \geq 0)$$

$$\text{Model II: } y_i^* = x_i'(\sqrt{2}\beta) + u_i, \quad u_i \sim \mathcal{N}(0,2), \quad y_i = 1(y_i^* \geq 0)$$

  leads to *identical* $\Pr(y \mid x)$.
- This is bad because the minimizer of $\mathrm{E}(-\log f(y \mid ; \theta))$ is not *unique*. Hence, MLE cannot be consistent.

Neglected heterogeneity

- Suppose the true underlying model is

$$\Pr(y = 1 \mid x, c) = \Phi(x'\beta + \gamma c) \tag{1}$$

  Assume $x \perp\!\!\!\perp c$ and $c \sim \mathcal{N}(0, \tau^2)$.

- For some reason, we neglect c and run probit with y on x only. What would be the consequence?

- We write eq. (1) in latent variable form as

$$y^* = x'\beta + \gamma c + e, \quad e \sim \mathcal{N}(0, 1), \quad y = 1(y^* > 0).$$

- Ignoring c would mean viewing $\gamma c + e$ as the "error term", and

$$\Pr(y = 1 \mid x) = \Pr(\gamma c + e > -x'\beta \mid x) = \Phi(x'\beta/\sigma),$$

  where $\sigma = \sqrt{\gamma^2\tau^2 + 1}$.

- Clearly, MLE ignoring c will converge to β/σ. Since, $\sigma > 1$, omitted variables create attenuation bias in probit model even when the omitted variable is *uncorrelated* with the included covariates.

- But $\hat{\beta}$ is still useful.

- First, we still can identify sign of the partial effect.

- Second, note that the true partial effect ratio is

$$\frac{\partial \Pr(y = 1 \mid x, c)/\partial x_j}{\partial \Pr(y = 1 \mid x, c)/\partial x_h} = \frac{\beta_j \phi(x'\beta + \gamma c)}{\beta_h \phi(x'\beta + \gamma c)} = \frac{\beta_j}{\beta_h}.$$

  This is identifiable since it equals $(\beta_j/\sigma)/(\beta_h/\sigma)$.

- The partial effect when $c = 0$ is not identifiable for any x:

$$\frac{\partial \Pr(y = 1 \mid x, c = 0)}{\partial x_j} = \beta_j \phi(x'\beta + 0).$$

- The following average partial effect is indeed identifiable,

$$\mathrm{E}_c \left(\left. \frac{\partial \Pr(y = 1 \mid x, c)}{\partial x_j} \right|_{x = x_0} \right). \tag{2}$$

We have

$$\begin{aligned}\mathrm{E}(y \mid x) &= \mathrm{E}(\mathrm{E}(y \mid x, c) \mid x) \\ &= \int \mathrm{E}(y \mid x, c) \, \mathrm{d}\, F(c).\end{aligned}$$

Thus,

$$\left.\frac{\partial \mathrm{E}(y \mid x)}{\partial x_j}\right|_{x=x_0} = \int \left.\frac{\partial \mathrm{E}(y \mid x, c)}{\partial x_j}\right|_{x=x_0} \mathrm{d}\, F(c) = \text{eq. (2)}$$

Obviously, $\partial \mathrm{E}(y \mid x)/\partial x_j = \beta_j/\sigma\phi(x'\beta/\sigma)$.

Continuous endogenous explanatory variables

- Consider the following model

$$y_1^* = z_1'\delta_1 + \alpha_1 y_2 + u_1 \tag{3}$$

$$y_2 = z_1'\delta_{21} + z_2'\delta_{22} + v_2 = z\delta_2 + v_2 \tag{4}$$

$$y_1 = 1(y_1^* > 0), \tag{5}$$

  where (u_1, v_2) has a zero mean, bivariate normal distribution, and is independent of z.

- Equation (3) along with eq. (5) is the structural equation; eq. (4) is called a reduced form for y_2. Since v_2 is normal, y_2 given z is normal. Hence y_2 should not be discrete.

- If u_1 and v_2 are correlated, y_2 is *endogenous*. If u_1 and v_2 are independent, there is no endogeneity problem.

- Question: when y_2 is endogenous, how to estimate δ_1 and α_1?

Rivers-Vuong's (1988) approach

- We normalize $\mathrm{Var}(u_1) = 1$ for identification.

- As for (u_1, v_2), we have the linear projection,

$$u_1 = \theta_1 v_2 + e_1,$$

  where $\theta_1 = \mathrm{Cov}(u_1, v_2)/\mathrm{Var}(v_2)$, and v_2 and e_1 is uncorrelated. Moreover, because (u_1, v_2) follows bivariate normal,

$$e_1 \sim \mathcal{N}(0, \mathrm{Var}(e_1)) \quad \mathrm{Var}(e_1) = \mathrm{Var}(u_1) - \mathrm{Corr}(u_1, v_2)^2 = 1 - \rho^2.$$

Here $\rho \equiv \mathrm{Corr}(u_1, v_2)$. We have

$$\rho = 1 - \theta_1^2 \,\mathrm{Var}(v_2).$$

Also note that v_2 and e_1 are indeed independent.

- We now can write

$$y_1^* = z_1\delta_1 + \alpha_1 y_2 + \theta_1 v_2 + e_1, \quad e_1 \sim \mathcal{N}(0, 1-\rho^2),$$

where e_1 is independent of z_1, y_2 and v_2.

- We have probit,

$$\Pr(y_1 = 1 \mid z, y_2, v_2) = \Phi\left(\frac{z_1\delta_1 + \alpha_1 y_2 + \theta_1 v_2}{\sqrt{1-\rho^2}}\right).$$

- If we know v_2, we can estimate

$$\delta_{\rho 1} = \frac{\delta_1}{\sqrt{1-\rho^2}}, \quad \alpha_{\rho 1} = \frac{\alpha_1}{\sqrt{1-\rho^2}}, \quad \theta_{\rho 1} = \frac{\theta_1}{\sqrt{1-\rho^2}}.$$

- Note that $\rho = 1 - \theta_1^2 \,\mathrm{Var}(v_2)$ and $\theta_{\rho 1} = \theta_1/\sqrt{1-\rho^2}$. We have

$$\frac{1}{1-\rho^2} = 1 + \theta_{\rho 1}^2 \,\mathrm{Var}(v_2),$$

hence

$$\text{Knowing } v_2 \Rightarrow (\delta_{\rho 1}, \alpha_{\rho 1}, \theta_{\rho 1}, \mathrm{Var}(v_2)) \Rightarrow (\delta_1, \alpha_1, \theta_1).$$

- Consistent estimates of v_2 can be obtained from the residuals $\hat{v}_2$ of the linear regression eq. (4).

- Here is a two-step procedure:

  1. Run the OLS regression y_2 on z and save the residuals $\hat{v}_2$;
  2. Run the probit y_1 on $z_1, y_2, \hat{v}_2$ to get consistent estimators of $\delta_{\rho 1}, \alpha_{\rho 1}, \theta_{\rho 1}$.

  Then then unscaled $\delta_1, \alpha_1, \theta$ is easily obtained.

- We can test the endogeneity of y_2 by testing $H_0 : \theta_1 = 0$ with simple t test. This is nice.

- The standard error can be easily derived using our two-step M estimation variance formula or bootstrap.

- Recall
$$y_1 = 1(y_1^* > 0), \quad y_1^* = z_1'\delta_1 + \alpha_1 y_2 + u_1, \quad u_1 \sim \mathcal{N}(0,1).$$

- We have estimated δ_1 and α_1.

- APE can be computed based on the difference and derivatives in $\Phi(z_1\hat{\delta}_1 + \hat{\alpha}_1 y_2)$.

- Consider y_1 as female labor force participation decision with $y_1 = 1$ (in labor force), and y_2 as husband's income. Imagine y_1^* as the net benefit of working, which depends on unobserved ability absorbed by u_1. We believe u_1 and y_1 are correlated. The difference,
$$\Phi\left(z_1\hat{\delta}_1 + \hat{\alpha}_1 \cdot (\bar{y} + 10,000)\right) - \Phi\left(z_1\hat{\delta}_1 + \hat{\alpha}_1 \cdot \bar{y}\right)$$
is the change of female LFP probability if husband's income increase by $10,000$ while other things (including the distribution of ability) unchanged. Causal effect!

- If we are only interested in the APE, we indeed do not need bivariate normal assumption. We have
$$\begin{aligned}\mathrm{E}(y_1 \mid z_1, y_2, v_2) &= \mathrm{E}\big(\mathrm{E}(y_1 \mid z_1, y_2, u_1, v_2) \,\big|\, z_1, y_2, v_2\big) \\ &= \mathrm{E}\big(\mathrm{E}(y_1 \mid z_1, y_2, u_1) \,\big|\, z_1, y_2, v_2\big) \\ &= \int \mathrm{E}(y_1 \mid z_1, y_2, u_1)\ \mathrm{d}\,F(u_1|v_2).\end{aligned}$$
Suppose we are interested in the partial effect of y_2, we have
$$\frac{\partial \mathrm{E}(y_1 \mid z_1, y_2, v_2)}{\partial y_2} = \int \frac{\partial \mathrm{E}(y_1 \mid z_1, y_2, u_1)}{\partial y_2}\ \mathrm{d}\,F(u_1 \mid v_2).$$
Now taking expectation of v_2 both sides, we have
$$\mathrm{E}_{v_2}\left(\frac{\partial \mathrm{E}(y_1 \mid z_1, y_2, v_2)}{\partial y_2}\right) = \int \frac{\partial \mathrm{E}(y_1 \mid z_1, y_2, u_1)}{\partial y_2}\ \mathrm{d}\,F(u_1).$$
Interchange the derivative and integral (expectation), we conclude that
$$\mathrm{E}_{v_2}\left(\frac{\partial \mathrm{E}(y_1 \mid z_1, y_2, v_2)}{\partial y_2}\right) = \frac{\partial \Phi(z_1\delta_1 + \alpha_1 y_2)}{\partial y_2}.$$
Note $\mathrm{E}(y_1 \mid z_1, y_2, v_2)$ is not a problem (nonparametrically estimable).

- The above approach is called control function approach.

- For a more general treatment of the endogeneity in binary choice model, Blundell and Powell (2004, *Review of Economic Studies*).

- For a general discussion about control function and its relationship with the IV, see Blundell and Powell (2003, "Endogeneity in nonparametric and semiparametric regression models" in *Advances in Economics and Econometrics*) and Imbens and Newey (2009, *Econometrica*).

- We can also estimate the above model using MLE, which turns out to be more efficient.

- We can decompose
$$f(y_1, y_2|z) = f(y_1|y_2, z)f(y_2|z).$$

- We have that $y_2|z \sim \mathcal{N}(z'\delta_2, \tau_2^2 \equiv \mathrm{Var}(v_2))$, and
$$\Pr(y_1 = 1 \mid y_2, z) = \Phi\left(\frac{z_1\delta_1 + \alpha_1 y_2 + (\rho_1/\tau_2)(y_2 - z'\delta_2)}{(1-\rho^2)^{1/2}}\right).$$
This formula was derived by the bivariate normal distribution assumption about (u_1, v_1).

- Given the formula of $f(y_1, y_2 \mid z)$, a (conditional) MLE can be applied to estimate the parameters (`'ivprobit'` in Stata).

- It is more efficient than the two-step approach whose second step is essentially about $f(y_1|y_2, z)$.

Binary endogenous explanatory variable

- Consider the following (triangle) model,
$$\begin{aligned} y_1 &= 1(z_1'\delta_1 + \alpha_1 y_2 + u_1 > 0) \\ y_2 &= 1(z'\delta_2 + v_2 > 0). \end{aligned}$$
When u_1 and v_2 are correlated, the binary y_2 is endogenous.

- The estimation relies on two results about bivariate normal. Let
$$(x_1, x_2)' \sim \mathcal{N}(0, \Sigma) \quad \Sigma = \begin{bmatrix} 1 & \rho \\ \rho & 1 \end{bmatrix}.$$
$$x_1|x_2 \sim \mathcal{N}(\rho x_2, 1-\rho^2), \quad f(x_1 = x|x_1 > -a) = \phi(x)/\Phi(a).$$

- The joint (conditional) likelihood is $\Pr(y_1, y_2 \mid z) = \Pr(y_2 \mid z)\Pr(y_1 \mid y_2, z)$.

- Let $a = (z_1'\delta_1 + \alpha_1 y_2 + \rho_1 v_2)/(1-\rho^2)^{1/2}$. Using the above facts about normal distributions,

$$\begin{aligned}
\Pr(y_1 = 1 \mid y_2 = 1, z) &= \mathrm{E}(y_1 \mid y_2 = 1, z) \\
&= \mathrm{E}(\mathrm{E}(y_1 \mid y_2 = 1, z, v_2) \mid y_2 = 1, z) \\
&= \mathrm{E}(\mathrm{E}(y_1 \mid z, v_2) \mid y_2 = 1, z) \\
&= \mathrm{E}(\Phi(a) \mid y_2 = 1, z) \\
&= \mathrm{E}(\Phi(a) \mid v_2 > -z\delta_2, z) \\
&= \Phi(z'\delta_2)^{-1} \int_{-z'\delta_2}^{\infty} \Phi(a)\phi(v_2)\ \mathrm{d}\, v_2.
\end{aligned}$$

- Similarly, we have

$$\Pr(y_1 = 1 \mid y_2 = 0, z) = [1 - \Phi(z'\delta_2)]^{-1} \int_{-\infty}^{-z'\delta_2} \Phi(a)\phi(v_2)\ \mathrm{d}\, v_2.$$

- `'biprobit'` in Stata.

5 Multinomial and ordered response models

Unordered and order outcomes

- Suppose the discrete outcome $y \in 0, 1, \ldots, J$.
- In the unordered (or nominal) case, the label (e.g. $1, 2, 3$) does not matter. For example, let y be the way of commuting. Suppose 0 means "driving", 1 means "walking", 2 means biking. Nothing essential changes if we switch the labels.
- Sometimes the order matters. For example, customer satisfaction may take values from 1 (terrible) to 5 (excellent). The fact that 3 is better than 2, and that 2 is better than 1 matters. Such outcomes are ordinal because the order has information.

Multinomial logit (MNL) model

- $y_i \in \{0, 1, \ldots, J\}$ is an *unordered* response. x_i changes by agent i but *not* alternative $0, 1, \ldots, J$.
- In the model of health insurance choice, x_i includes observable characteristics of individual i but not of different kinds of insurance plans.

- This is obviously limited, since the features, such as premium, of each insurance plan also affect the choice probability.

- CCP: $p_j(x) = \Pr(y = j \mid x)$ for $j = 0, \ldots, J$. Of course, $\sum_{j=0}^{J} p_j(x) = 1$. We are interested in $\partial p_j(x)/\partial x_\ell$.

- The MNL model:

$$p_j(x) = \frac{\exp(x'\beta_j)}{1 + \sum_{h=1}^{J} \exp(x'\beta_h)}, \quad j = 1, \ldots, J,$$
$$p_0(x) = \frac{1}{1 + \sum_{h=1}^{J} \exp(x'\beta_h)}.$$

  Note that β_j changes with alternative.

- Example: Keane and Wolpin (1997, *Journal of Political Economy*).

- A young men i at age 16 chooses his or her career among five mutually exclusive and exhaustive alternatives: work in either a blue- or white-collar occupation (1), work in the military (2) attend school (3), or engage in home production (0).

- x_i could include "years of education", "working experience", "working experience squared", "race dummy".

- β_j varies across alternatives, because the return to these x_i is different across occupation.

- The partial effect in MNL is complicated,

$$\frac{\partial p_j(x)}{\partial x_k} = p_j(x) \left(\beta_{jk} - \frac{\sum_{h=1}^{J} \beta_{hk} \exp(x'\beta_h)}{1 + \sum_{h=1}^{J} \exp(x'\beta_h)} \right).$$

- Take a look again:

$$\frac{\partial p_j(x)}{\partial x_k} = p_j(x) \left(\beta_{jk} - \sum_{h=1}^{J} \beta_{hk} p_h(x) \right).$$

- Odds ratio is easier to interpret:

$$\frac{p_j(x)}{p_0(x)} = \exp(x'\beta_j).$$

  Log odds of choosing j relative to the reference alternative 0 is

$$\log \left(\frac{p_j(x)}{p_0(x)} \right) = x'\beta_j.$$

  β_{jk} is the partial effect of x_k on the log odds of j relative to 0.

- MLE of β_j is simple. The log likelihood for $z_i = (x_i, y_i)$ is

$$\ell_i(\beta) = \sum_{j=0}^{J} 1(y_i = j) \log p_j(x_i; \beta).$$

  Inference is also standard.

McFadden's Probabilistic choice models

- Let's use latent variable form again. View y_{ij}^* as i's latent utility of choosing alternative j,

$$y_{ij}^* = x_{ij}'\beta + a_{ij}, \quad j = 0, \ldots, J,$$

  where x_{ij} can vary by individual i and alternative j. Now β does not change with j.

- In the health insurance plan choice, x_{ij} can include the premium of each plan (does not change across individual i) and i's demographic characteristics (age, sex et al.), which does not change across plan (j).

- Let x_i include all non-redundant x of $(x_{i0}, x_{i1}, \ldots, x_{iJ})$. Let $a_i = (a_{i0}, \ldots, a_{iJ})$ and assume $a_i \perp\!\!\!\perp x_i$.

- The choice y_i was generated from

$$y_i = \arg \max_{j \in \{0,1,\ldots,J\}} y_j^*$$

  to maximize the utility.

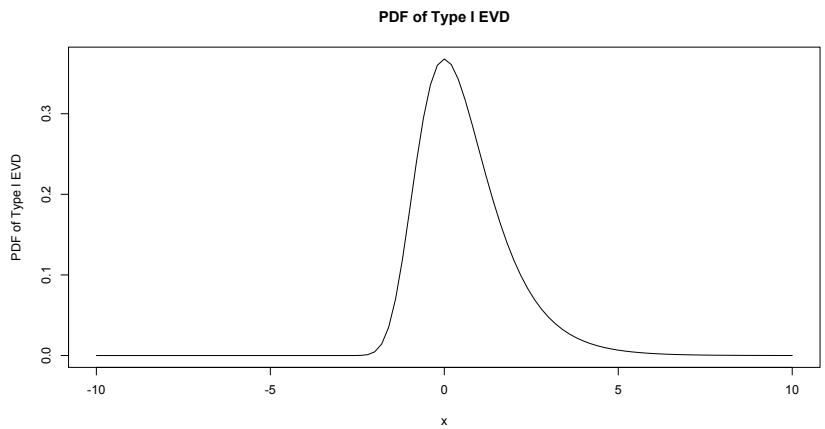
- If $\{a_{ij} : j = 0, 1, \ldots, J\}$ are independent, identically distributed with type I extreme value distribution(EVD) ($F(a) = \exp(-\exp(-a))$), then

$$\Pr(y_i = j \mid x_i) = \frac{\exp(x_{ij}'\beta)}{1 + \sum_{h=1}^{J} \exp(x_{ih}'\beta)}.$$

  It is usually called *conditional logit model*.

- The MLE estimation of β is straightforward.

- The assumption about type I EVD is a little bit odd. It is not symmetric and has a thicker right tail. Its mean equals approximately 0.5772 (Euler's constant).



- MNL can be viewed as a special case of conditional logit:

$$\text{MNL:} \quad \Pr(y_i = j \mid w_i) = \frac{\exp(w_i'\alpha_j)}{1 + \sum_{h=1}^{J} \exp(w_i'\alpha_h)}.$$

  Write $x_{ij} = (d1_j w_i, d2_j w_i, \ldots, dJ_j w_i)'$, where $dk_j = 1$ if $k = j$ and 0 otherwise. Let $\beta = (\alpha_1, \ldots, \alpha_J)$. We have then

$$x_{ij}'\beta = w_i'\alpha_j \quad j = 1, \ldots, J.$$

- Then the conditional logit with the above defined x_{ij} and β gives the same choice probabilities as MNL does.

- In many applications, the utility specification is as follows,

$$y_{ij}^* = z_{ij}\gamma + w_i\delta_j + a_{ij}, j = 0, 1, \ldots, J,$$

  with $\delta_0 = 0$. This is called mixed logit model.

- This fits $x_{ij}'\beta$ specification by letting,

$$x_{ij} = (z_{i0}', \ldots, z_{iJ}', 1_J' \otimes w_i')'$$
$$\beta = (1_{J+1}' \otimes \gamma', \delta_1', \ldots, \delta_J')'$$

IIA of conditional logit

- A key restriction of conditional logit model is independence from irrelevant alternatives (IIA). It is easy to see that

$$\frac{p_j(x_{ij})}{p_h(x_{ih})} = \frac{\exp(x_{ij}'\beta)}{\exp(x_{ih}'\beta)} = \exp((x_{ij} - x_{ih})'\beta)$$

- This means the relative probabilities for any two alternatives depend *only* on the attributes of those two alternatives.
- It is called IIA because adding another alternative or changing the characteristics of a third alternative does not affect the relative odds between j and h, even when the new alternative is relevant to the existing ones.
- This is not desirable in some applications.
- The most well known example illustrating the limitation of the IIA is called red-bus-blue-bus by McFadden.

- The choice is about the mode of transportation. Initially, there are two alternatives "car" and "red bus". Suppose a commuter chooses between them with equal probability 1/2.
- Now we have another alternative "blue bus". Suppose commuters do not care about color. Then by IIA, the relative odds between "car" and "red bus" is one. Moreover, the relative odds between "car" and "blue bus" is also one. Hence, the probability of choosing any alternative is 1/3.
- This means the fraction of commuters taking a car would fall from 1/2 to 1/3.
- It is unrealistic. To a commuter who does not care about color, a choice among "car", "red bus", "blue bus" is more or less like choice between "car" and "bus". We expect the fraction of car commuting does not change!
- Consider another example in the industrial organization.
- View the two alternatives j and h are two car models, e.g. j is Toyota Yaris (economy car) and h is BMW 5 series (luxury type). The choice probability represents the market share each car model.

- Suppose x_{ij} and x_{ih} include price. IIA then implies that the price elasticity of relative market share will *not* be affected by the third car model!
- This is unrealistic. For example, if the price of a Mercedes E series (competitor of BMW 5 series) drops, we expect that a part of consumers might choose Mercedes over BMW. Hence the relative market share between Yaris and BMW 5 will increase.

Relax IIA

- In the red-blue bus example or car example, we assumed the latent utility

$$y_j^* = \mu_j + a_j, \quad j = 0, 1, 2,$$

  and a_j are independent of each other.
- This does not make sense. We expect the utility shocks of red and blue buses (BMW and Mercedes) are correlated.
- IIA results from the independence assumption. Nested logit model is to impose a different distribution about $a_0, a_1, \ldots, a_J$ such that certain correlation is allowed.
- Let 0 be "car", 1 be "red bus", 2 be "blue bus". Assume

$$F(a_1, a_2) = \exp\left(-\left[\exp(-\rho^{-1}a_1) + \exp(-\rho^{-1}a_2)\right]^{\rho}\right), \quad 0 < \rho \leq 1.$$

  It can be shown that $\mathrm{Corr}(a_1, a_2) = 1 - \rho^2$. If $\rho = 1$, a_1 and a_2 are independent.
- Assume a_0 follows the standard type I EVD.
- Under these assumptions, it can be shown that (see section 9.3.5 of Amemiya: *Advanced Econometrics*),

$$\Pr(y = 0) = \frac{\exp(\mu_0)}{\exp \mu_0 + \underbrace{[\exp(\rho^{-1}\mu_1) + \exp(\rho^{-1}\mu_2)]^{\rho}}_{\text{weighted average of } \exp(\mu_1) \text{ and } \exp(\mu_2)}} \tag{6}$$

$$\Pr(y = 1 \mid y \neq 0) = \frac{\exp(\rho^{-1}\mu_1)}{\exp(\rho^{-1}\mu_1) + \exp(\rho^{-1}\mu_2)}. \tag{7}$$

- Equation (7) shows that the choice between the two *similar* alternatives is made according to a binary logit model.
- Equation (6) shows the choice between car and non-car (a distinct group) is also like a logit model. The utility from non-car alternative is a certain kind of weighted average of the utilities from alternatives in the non-car group.

- Suppose we can group alternatives intro S groups of "similar" alternatives.

$$(0, 1, 2, \ldots, J) = B_1 \cup B_2 \cup \cdots \cup B_S.$$

We can use the following joint distribution

$$F(a_0, \ldots, a_J) = \exp\left(-\sum_{s=1}^{S} \alpha_s \left[\sum_{j \in B_s} \exp(-\rho_s^{-1} a_j)\right]^{\rho_s}\right).$$

- It can be shown that,

$$\Pr(y \in B_s \mid x) = \frac{\alpha_s (\sum_{j \in B_s} \exp(\rho_s^{-1} x_j' \beta))^{\rho_s}}{\sum_{r=1}^{S} \alpha_r (\sum_{j \in B_r} \exp(\rho_r^{-1} x_j' \beta))^{\rho_r}}, \tag{8}$$

$$\Pr(y = j \mid y \in B_s, x) = \frac{\exp(\rho_s^{-1} x_j' \beta)}{\sum_{h \in B_s} \exp(\rho_s^{-1} x_h' \beta)}.$$

- MLE can of course be done. Here is a simple two-step approach.

  1. Estimate β/ρ_s using standard logit for each group B_s.
  2. Insert the estimates of β/ρ_s for $s = 1, \ldots, S$ into eq. (8) and estimate α_s and ρ_s.

- Stata command `'nlogit'`.

Ordered probit model

- Now consider the ordered response case. Suppose $y \in \{0, 1, \ldots, J\}$.

- The ordered probit model assumes that the latent variable

$$y^* = x'\beta + \varepsilon, \quad \varepsilon | x \sim \mathcal{N}(0, 1).$$

In this model, x does not include a constant. Let $a_1 < a_2 < \cdots < a_J$ be J unknown cut points. We will estimate a and β.

- Assume that

$$\begin{aligned} y &= 0 \text{ if } y^* \leq a_1 \\ y &= 1 \text{ if } a_1 < y^* \leq a_2 \\ &\vdots \\ y &= J-1 \text{ if } a_{J-1} < y^* \leq a_J \\ y &= J \text{ if } y^* > a_J \end{aligned}$$

- We have

$$
\begin{aligned}
\Pr(y = 0 \mid x) &= \Phi(a_1 - x'\beta) \\
\Pr(y = 1 \mid x) &= \Phi(a_2 - x'\beta) - \Phi(a_1 - x'\beta) \\
&\vdots \\
\Pr(y = J-1 \mid x) &= \Phi(a_J - x'\beta) - \Phi(a_{J-1} - x'\beta) \\
\Pr(y = J \mid x) &= 1 - \Phi(a_J - x'\beta)
\end{aligned}
$$

- If x includes constant, a would not be identified.
- Given the probabilities, the MLE and inference is standard.